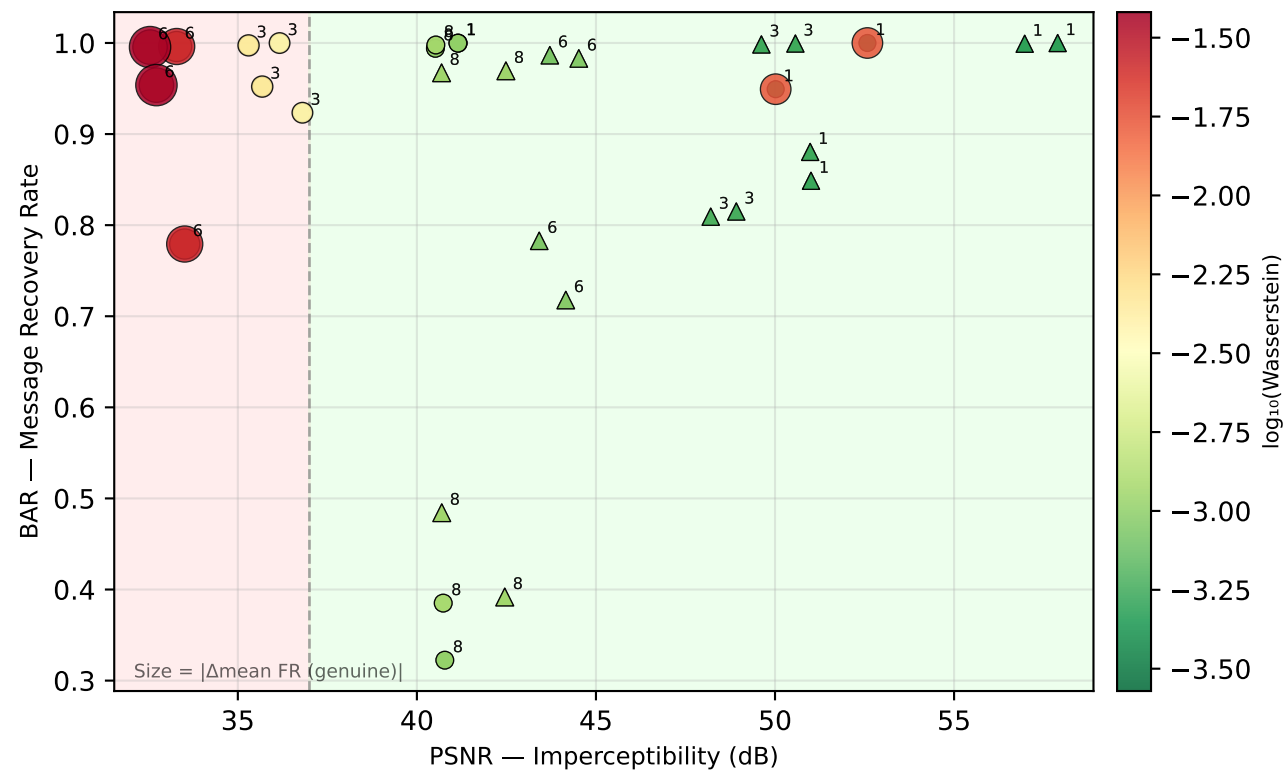
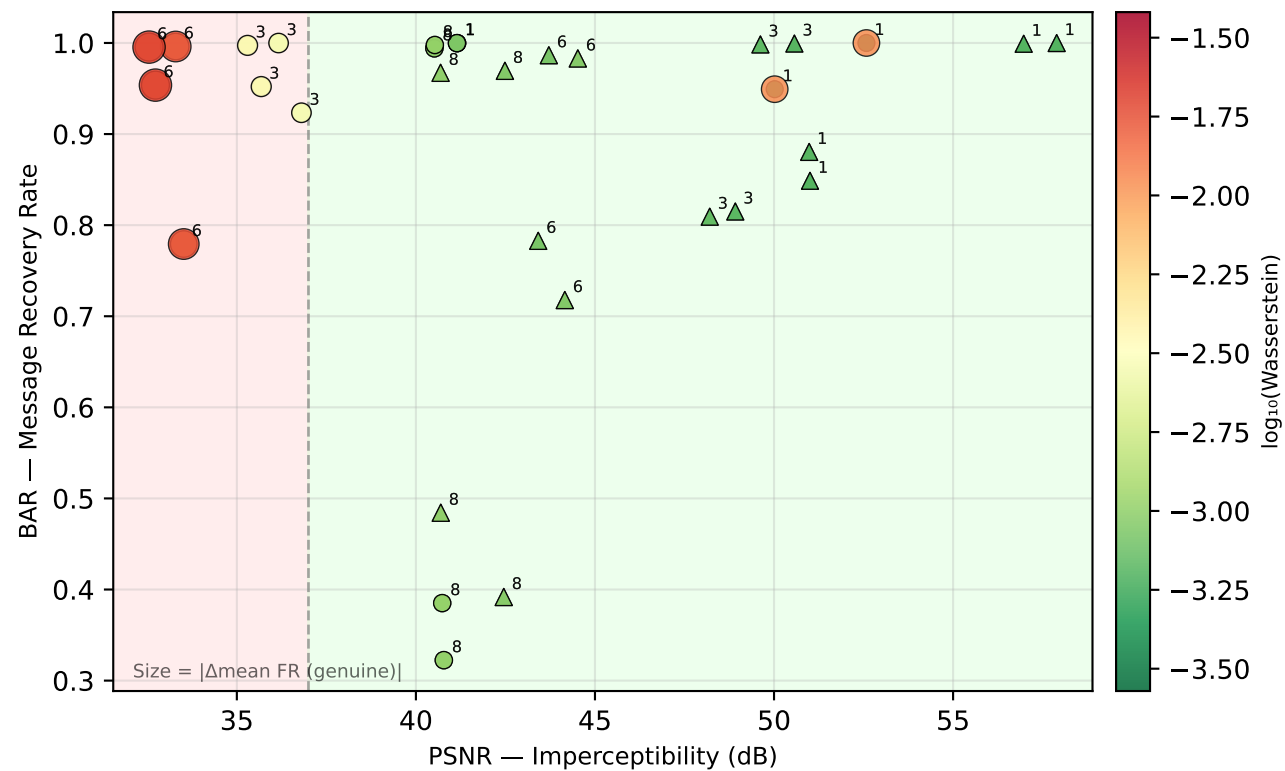


Pareto Frontier: PSNR vs Recovery | Color = $\log_{10}(\text{Wasserstein})$ | Size = $|\Delta\text{mean FR}|$
All Conditions Together — Genuine Pairs

Arcface — All Conditions



Facenet — All Conditions



Watermarking Algorithm

▲ Stegaformer

● Stegformer

PSNR < 37.0 dB — FR affected

PSNR ≥ 37.0 dB — FR safe